

• Monty Python / Shakespeare
→ robots / entering data

- Code quality

↳ a concern that should
be going away?

[D] IDE + training has
always been there

- data quality

→ LLM inference

only starting to
scratch the surface

- data reliability

→ Toner, Toronto Women,
Gino

→ robots

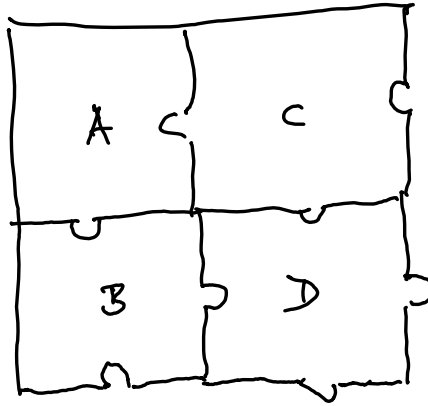
- execution trustworthiness

When things fail

→ retractions in
PubMed

Credibility of economic research

4 pillars and thoughtlets on future paths



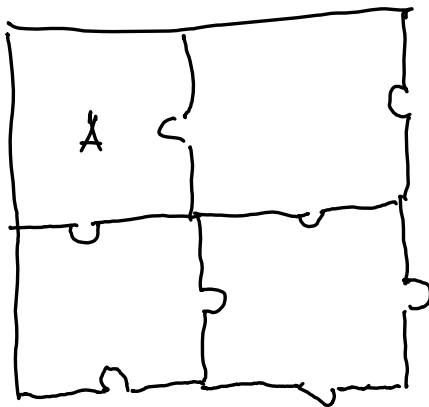
A: Data exists, and is of sufficient quality

B: Data provenance: trustworthy, open/
accessible

C: Code: exists, works, is reusable, comprehensible

D: Documentation: Paper, Readme, actionability
info

data quality →



- Administrative data
- High quality survey data

↳ Survey data traditionally well documented (methods report)

Admin data "fit for purpose"? Depends on usage, coverage, ultimately expert assessment

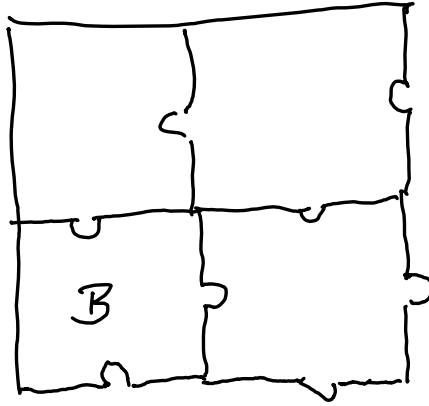
- often thinner, longer

Business data: fine detail, but possibly non-representative

Challenges [Part 2, recall]

- UN-generated data popular
 - draw from inference distribution
 - variability important (no recent paper)
 - treatment needs to expand (my 41 additions)

Data provenance →



Part of data quality is provenance

→ monkey picture

Gold standard: - statistical agency surveys
- big research institutions

Gold standard: DOI assignment, citation
→ mostly absent in statistical agencies
[show 404 rings by government]
[show Data Rescue website]

But: concerns (traditional)

- Argentinean inflation data

Query:

Do you

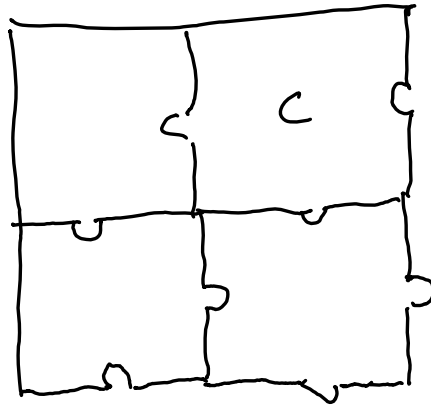
know

(w/ name)

[- Gino data falsification
- Psycho Guy

⇒ point to tutorial

[Part 2]: Tower: LFM inverted data?



Code

Statistics / Analysis:

- How to capture code quality
- Computer Science: Efficiency, coverage
- Rubric (show)
 - Distribution slide
- Reproducibility statistics
 - Number of reports w/

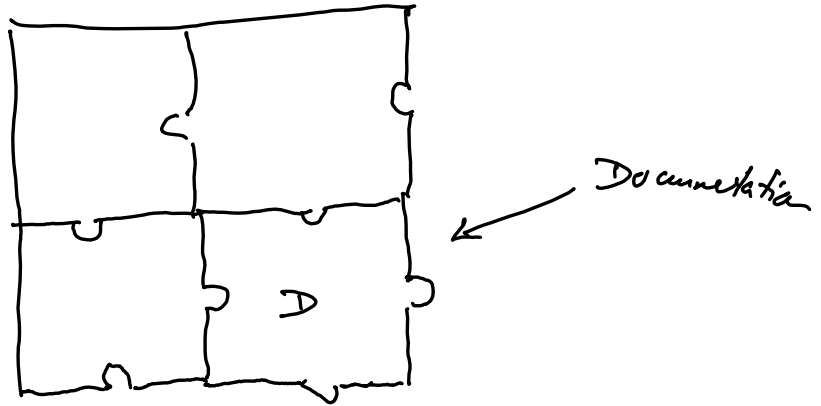
- Bugs in code
- Code not functioning
- Code missing
- Discrepancy

Jira
query

- Same from World Bank
(blog post)

could
data availability
(excl. cases w/
data not available)

- Statistics from i42
- Connect to data availability
(insert slides)



Statistics:

- from R&P 2026: split across categories
- from Sebastian: utility over time
- from Carolyn paper: discrepancy between AEA papers and others

Issues

- too brief
- insider knowledge
- not documented in Article, but no other place

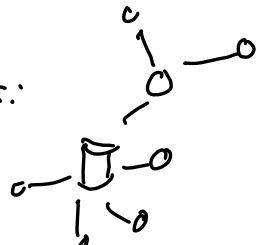
Thoughts: What can researchers do?

Data quality: - As (co-) data producers:

- Quality reports

- Fit for purpose:

Same tool as for literature
Data citation allows for tracking, assessment



Data provenance: - Evidence: DOI → checksum → verifiable

- Testimony: Data Editor serving as verifier

→ image of title page footnote

→ also limits

(Chinese mobile network)

- Transparency!

[Side note: LLM provenance, evidence of data quality] → variability

LLM training materials

→ example historical records - Clemons

- Dell

Code quality

Import of AI

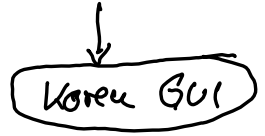
- should improve

- but: complexity (increased presence of increased scaffolding takefile, Python, Linux)

- AI in secure environments?

Documentation

- AI can help : consistency betw/ article + code
- Readme template can help
 - feed it to AI



But:

- please tell it to be brief
- no CoR
- Re-read : you are still responsible